HW 12 — GenAI Audit Log (Final Project Capstone)

Student: Jess Barrett, EID 854433853
Course: CIS 530 — Spring 2026
Project: Comparing DQN and Double DQN for Spacecraft Landing Control in LunarLander-v3
Repository: https://github.com/jess-barrett/LunarLander-DQN
LLM Used: Claude (Anthropic, claude.ai)
GenAI Kit Version Followed: GenAI Kit v3 — Comprehensive Project Guardrails
Submission Date: May 4, 2026

Declaration of API Quotas / Compute Limits

This project used Anthropic's Claude (claude.ai web interface) under a standard consumer subscription. No API quotas, paid compute, or specialized "Exoskeleton" workflows beyond the standard chat interface were utilized. Code execution was performed in Google Colab on a free-tier T4 GPU runtime.

Originality Statement

The core ideas, project topic selection (LunarLander-v3 + DQN/DDQN comparison), experimental design choices (5 trials, paired evaluation seeds, statistical hypothesis formulation), and interpretation of empirical findings are my own. The LLM was used to:

- Synthesize my raw bullet points into formal academic prose under the "Grammarly on Steroids" policy.
- Produce code skeletons (DQN architecture, experiment harness, analysis pipeline) which I executed, validated, and own as my project artifacts.
- Reformat my drafts into IEEE LaTeX template structure.
- Critique and structure my slide deck outline.

The LLM was *not* used to invent results, fabricate citations, or generate methodology decisions independent of my input.

Section 1 — Phase 1 (Architecture & Data Pipeline)

The Phase 1 architecture critique prompt was not used in its formal template form for this project. Instead, project methodology was developed iteratively earlier in the semester (PS4 proposal, PS8 interim report) and locked in before HW12. The final architecture (DQN vs DDQN with paired evaluation seeds) was decided by the student based on the available rubric requirements.

Section 2 — Phase 2 (Debugging & Implementation)

2.1 Initial Debugging — Local Python Environment

ISSUE ENCOUNTERED

> Python 3.14 on Windows could not install pygame as a dependency of gymnasium[box2d]. Error: "ModuleNotFoundError: No module named 'setuptools._distutils.msvccompiler'" during pip install.

AI RESPONSE (LOCALIZED FIX PROVIDED)

> Recommended switching to Google Colab to avoid Windows-specific build chain incompatibility with Python 3.14. Did not rewrite the script. Provided the conceptual cause (newer Python lacks compiled wheels for pygame) and the localized fix (use Colab where dependencies are pre-resolved).

2.2 Gymnasium API Version Mismatch

ISSUE ENCOUNTERED

> DeprecatedEnv: Environment version v2 for `LunarLander` is deprecated. Please use `LunarLander-v3` instead.

AI RESPONSE (LOCALIZED FIX PROVIDED)

> Single-line conceptual explanation (Gymnasium dropped v2 in latest release) and a one-line fix (change "LunarLander-v2" to "LunarLander-v3" in env.make calls). No rewrite of unrelated code.

Section 3 — Phase 3 (Evaluation Strategy)

PROMPT USED (PARAPHRASED)

```
My model outputs continuous reward values from a reinforcement learning agent across multiple training trials, comparing DQN to DDQN on LunarLander. What is the most appropriate statistical significance test for this data type? What metrics should I visualize? What methodological flaw would cause you to reject my evaluation?
```

ORIGINAL NOTES PROVIDED

> - 5 trials per method - Each trial evaluated over 100 fixed episodes - Same evaluation seeds across methods (so trials are paired) - Reward values are continuous, not bounded normal - Want to compare DQN vs DDQN and both vs random baseline

AI SUGGESTIONS (SYNTHESIZED)

> - Paired t-test as primary inferential test (paired trials share evaluation seeds)
> - Wilcoxon signed-rank test as non-parametric sanity check (no normality assumption)
> - Cohen's d effect size to be reported alongside p-values, since deep RL has well-documented high inter-trial variance
> - Visualizations: training curves (mean ± std), evaluation reward distributions (box plot), per-trial means (bar chart)
> - Methodological flaw flagged: with N=5 paired observations, Wilcoxon's minimum achievable p-value is 0.0625; t-test must be primary

Section 4 — Final Phase: Report Editor (v3)

4.1 System Prompt Used

```
Act as a strict academic editor for a graduate-level Artificial Intelligence course. I am writing my final term project report. YOUR RULES: 1. You must NOT generate novel ideas, architectures, implementation details, or results. 2. If I ask you to write a section, you must REVERSE-PROMPT me by asking targeted questions about my methodology, dataset, and empirical findings. Only once I provide my raw notes or code may you summarize and refine my writing. 3. Enforce the grading rubric: ensure I have clearly defined my Problem Formulation, Methodology, and Statistical Evaluation. 4. Ensure my tone matches formal IEEE/ACM conference proceedings. Keep sentences tight and academic.
```

4.2 Reverse-Prompted Discussion Section

AI REVERSE-PROMPTS

1. Looking at DQN trial evaluation means (94, 162, 204, 126, 171), are you surprised by the variance between trials?
2. Which strikes you as the most interesting story in the data?
3. Going in, what was your prediction for DQN vs DDQN?
4. Do you think the DQN variance suggests some trials needed MORE training episodes?
5. Were DQN and DDQN trained under truly identical conditions?
6. Which evaluation metric do you think tells the most honest story for RL: p-value or Cohen's d?

STUDENT'S ORIGINAL ANSWERS (RAW)

1. Yes, very surprising — all of those things surprised me (range from 94 to 204, only one trial hit "solved", trial 1 was way below others)
2. Whether DDQN actually beats DQN
3. DDQN should win because of overestimation bias — that's the whole point of the algorithm
4. Maybe — not sure if 5 trials is enough to tell
5. Yes — same architecture, hyperparameters, episode count, and evaluation seeds
6. Cohen's d (effect size) — because RL has so much variance, we want to know how BIG the difference is

SYNTHESIZED OUTPUT (DISCUSSION SECTION, FINAL REPORT)

> "The more interesting result is the comparison between DQN and DDQN. Even though DDQN is supposed to reduce overestimation bias (van Hasselt et al., AAAI 2016), there was no statistically significant improvement over vanilla DQN ($t = 0.83$, $p = 0.46$). The point estimate actually slightly favored DQN (mean 151.5 vs 117.6), with a small negative effect size ($d = -0.53$) for DDQN. Two things probably explain this. First, the variance across DQN trials was already pretty wild (94.3 to 204.0 — a range over 100 reward points for the same algorithm with only the random seed varying), and DDQN's variance was even higher (5.4 to 191.6). With only $N = 5$ trials, the comparison is underpowered: a single unlucky seed can shift the mean by tens of points. Second, the environment may not really show off DDQN's strengths — overestimation bias scales with the size of the action set, and LunarLander-v3 only has four discrete actions..."

4.3 Phase 2 Synthesis Prompt (Used Earlier in PS8)

ORIGINAL SYNTHESIS PROMPT

```
[SYSTEM INSTRUCTION] You are an expert technical academic editor. I am providing my original bullet points, raw data, and architectural choices below. Please synthesize this into a formal, concise academic response. Do NOT add new claims, invent data, hallucinate citations, or alter my original intellectual intent.

[MY ORIGINAL INPUT]
Track: 1 (Search, Games, & Constraint Satisfaction)
Pillar: RL (Reinforcement Learning)
```

```
My Raw Notes/Data:
- Training a DQN agent to land a spacecraft in LunarLander-v3
- State space is 8 continuous dimensions, 4 discrete actions
- Random baseline: mean reward of -169.58 over 100 episodes
- Q-network: two hidden layers of 128 ReLU units
- Target network syncs every 10 episodes
- Experience replay buffer: 100k transitions, batches of 64
- Epsilon-greedy: 1.0 → 0.01, decay 0.995/episode
- Adam optimizer, lr=1e-3, MSE loss
- Sources: Mnih 2013 arXiv, Mnih 2015 Nature, Sutton & Barto Ch. 6
- Chose DQN over tabular Q-learning (continuous state space)
- Chose DQN over PPO/A2C (well-documented for this env)
```

Section 5 — Final Phase: Presentation Formatter (v3)

5.1 System Prompt Used

```
Act as a presentation coach. Review my slide deck outline against the following strict course requirements: 1. Total slide count must be 10-15 for a solo project. Target 60-90 seconds per slide. 2. Check that I have exactly ONE slide for each of the three standard HW12 reflection questions, and that each team member has a dedicated bullet point on those slides. 3. Verify that I am using inline APA-like citations (truncated author names with venue), NOT numbered citations, and that references are in footnotes. 4. Remind me to check that no font on my slides is smaller than 18pt (except footnotes/captions) and that all chart axes are labeled. 5. Ask me if my code snippets are inline rather than screenshots, and if my screenshots replace live code walkthroughs.
```

5.2 Student's Reflection Question Answers (Raw)

Q1 (Most challenging): Understanding why DDQN didn't beat DQN despite the theory.
Q2 (Learned the most about): How DQN works internally — replay buffer, target network, Bellman targets.
Q3 (Top future work): Run more trials (N ≥ 30) for tight confidence intervals on DQN vs DDQN.

5.3 Compliance Checklist (verified before deck export)

Requirement	Status
Slide count 10–15 for solo (excl. cover)	✅ 15 + cover
One slide per reflection question	✅ Slides 12, 13, 14
Inline APA-style citations, not numbered	✅ "(Mnih et al., Nature 2015)" format
References in footnotes	✅ Per-slide footnotes
No font below 18pt (except footnotes)	✅ Body text 18pt+
Chart axes labeled	✅ Generated via analysis.py
Code inline, not screenshots	✅ Equations only, no code blocks

Section 6 — Verification Statement

All raw notes, observations, hypotheses, and interpretations were authored by the student before being provided to the AI for editing or formatting. The AI was used as a synthesizer of student-provided content and a structural editor for IEEE LaTeX formatting, not as a generator of substantive ideas. All citations were manually verified to be real and accurate:

- Mnih et al. 2013 — verified at arXiv:1312.5602
- Mnih et al. 2015 — verified at Nature Vol. 518, pp. 529–533
- van Hasselt et al. 2016 — verified at AAAI Proceedings, Vol. 30
- Sutton & Barto 2018 — verified MIT Press 2nd edition
- Henderson et al. 2018 — verified at AAAI Proceedings, Vol. 32
- Brockman et al. 2016 — verified at arXiv:1606.01540

All empirical numbers reported in the final paper (means, standard deviations, t-statistics, p-values, Cohen's d values) were generated by the student's own code (experiment.py, analysis.py) running in Google Colab and saved to experiment_results.npz. The AI did not fabricate any results.

> **Signature:** Jess Barrett, May 4, 2026
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